Sahaj Bhadja

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Summary

Backend software engineer with 2+ years building **reliable Python/Node.js services**, REST APIs, transaction workflows, and distributed data systems; shipped tested platforms handling 2M+ records, 1M+ transactions, and 10K+ daily requests across AWS and Azure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python FastAPI services and normalized schemas** for 500+ users, implementing row-level isolation, indexed queries, and optimized access patterns that reduced analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Developed **fault-tolerant data pipelines** processing 5,000+ monthly jobs with incremental loads, reconciliation, validation, error handling, and CI/CD releases, achieving 99%+ production reliability across recurring distributed workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **Python Flask backend** serving 150+ users and 10,000+ daily requests, implementing concurrent batch execution and fault-tolerant APIs to process 1,000+ files while maintaining stable peak-load performance.
- Owned features from **product requirements through deployment**, delivering secure ingestion, validation, persistent results, usage logging, fallback handling, and GitHub Actions releases while reducing user feedback turnaround by 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Troubleshot and optimized **API and ETL integrations** using OOP, AWS Glue, S3, SQL, caching, JSON, and Parquet transformations, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Automated **builds, deployments, and 20+ unit/integration suites** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while reducing regressions and technical debt 20% across releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **Node.js APIs across eight microservices** for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through WebSockets and Azure SQL while reducing cross-location sync conflicts 33%.
- Implemented **JWT authorization, automated testing, and observability** with Jest, Selenium, Sentry, Azure Monitor, and Kibana, then packaged services through Docker and Jenkins CI/CD, cutting release time 50%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Developed **Python Flask APIs for transaction workflows** covering bookings, pricing, payments, refunds, and webhook-driven status updates, maintaining sub-200-millisecond responses across 50,000+ records through real-time Firebase synchronization.
- Integrated **multi-currency payment and reconciliation workflows** across cards, wallets, refunds, and disputes, reducing manual payment handling 70% while maintaining consistent transaction state through automated idempotent webhook processing.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, distributed systems, concurrency
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase, ETL, indexing, reconciliation
Reliability: PyTest, Mockito, Jest, Selenium, Sentry, Azure Monitor, Kibana, error handling, performance tuning
Platform: AWS, Azure, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*
- Built a **financial data platform** supporting 50,000+ users and 1M+ transactions, integrating Plaid for secure bank linking, real-time synchronization, advanced filtering, MongoDB access, and Kubernetes-based CI/CD automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered data platform** supporting 10+ SQL and NoSQL systems, accelerating query authoring, schema exploration, and data management through a cross-platform Java and TypeScript codebase with Docker workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python data-processing API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.